



🕒 Time: 5 minutes

🕒 Outcome: Calculate Average Costs: AFC, AVC & ATC.

🕒 Difficulty: Intermediate

## Average Costs: AFC, AVC & ATC



### THE CORE IDEA

Average fixed cost, average variable cost, and average total cost express costs per unit of output. Average total cost equals average fixed cost plus average variable cost. Average fixed cost falls as output spreads the same fixed cost across more units, while average variable cost changes with the productivity and price of variable inputs.



### HOW TO RECOGNIZE IT

- Divide fixed cost by output for average fixed cost and variable cost by output for average variable cost.
- Average total cost equals average fixed cost plus average variable cost.
- Average fixed cost falls continuously as output spreads the same fixed cost across more units.
- Use per-unit measures only when output is positive.

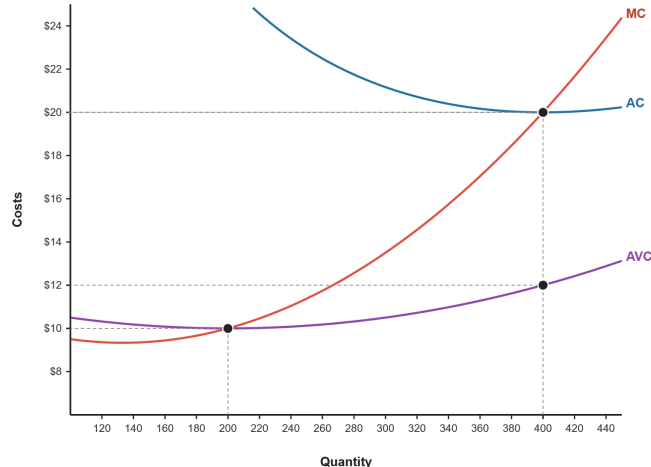


### WATCH OUT

Average costs are per unit; total costs are not. Subtract AVC from ATC to find AFC, then multiply AFC by positive output to recover total fixed cost.



### WORKED EXAMPLE: READING AVERAGE-COST CURVES



At 400 units, the graph shows average cost of \$20 and average variable cost of \$12. Because  $AC = AVC + AFC$ , average fixed cost is  $\$20 - \$12 = \$8$  per unit. The gap between AC and AVC is therefore the firm's average fixed cost.



### CHECK YOURSELF

At  $Q = 30$ , AVC is \$14 and ATC is \$24. Calculate AFC and total fixed cost. Verify your answer by multiplying the per-unit gap by output.



**READY?**

**Return to the game and master this concept.**